Daily Practice

Chapter: Unit 1: Limits & Continuity
Recommended Time: 45 Minutes

Part I: Essential Concepts & Formulas ( )

1. Existence of a Limit ( )

For a limit to exist at $x = c$,

$$\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$$

Note: *(absolute values)* *(piecewise functions)*

2. Evaluating Limits Analytically ( )

(Direct Substitution) $\frac{0}{0}$ (Indeterminate Form)

1. **Factoring ( ):** 0
2. **Conjugates ( ):** ($\sqrt{x} - a$ $\sqrt{x} + a$)

3. Special Trigonometric Limits & Equivalent Infinitesimals ( )

(Squeeze Theorem)

- $\lim_{x \to 0} \frac{\sin x}{x} = 1$
- $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$

Equivalent Infinitesimals ( ): AP $x \to 0$

- $\sin x \approx x$
- $\tan x \approx x$
- $1 - \cos x \approx \frac{1}{2}x^2$

4. Asymptotes & Infinite Limits ( )

Horizontal Asymptotes (HA) / : $\lim_{x\to\infty} f(x)$ $\lim_{x\to-\infty} f(x)$
(highest degrees)

- **Bottom Heavy ( ):** $0 \implies$ HA is $y = 0$.
- **Top Heavy ( ):** $(\pm\infty) \implies$ No HA.
- **Degrees are Equal ( ):** (Divide the leading coefficients).
- **(The Negative Infinity Trap):** $\lim_{x\to-\infty}$
  $\sqrt{x^2} = |x|$ x $|x| = -x$

Vertical Asymptotes (VA) / : x 0 0
VA 0 Hole

5. Continuity Checklist ( )

To prove a function $f(x)$ is continuous at $x = c$, you MUST show all 3 steps on the FRQ section ( ):

1. $f(c)$ exists ( ).
2. $\lim_{x\to c} f(x)$ exists ( ).
3. $\lim_{x\to c} f(x) = f(c)$ ( ).

6. Intermediate Value Theorem, IVT ( )

AP Standard Justification Template :
"Because $f(x)$ is continuous on the closed interval $[a,b]$, and $f(a) = A$, $f(b) = B$. Since k is between A and B, by the Intermediate Value Theorem (IVT), there must exist a value c in the interval (a,b) such that $f(c) = k$."

Part II: AP Practice

Section 1: Multiple Choice Questions

No Calculator Allowed.

MCQ 1. What is the value of $\lim_{x\to4} \frac{x^2-16}{x^2-5x+4}$?

A) 0
B) $\frac{8}{3}$
C) $\frac{4}{5}$
D) The limit does not exist.

MCQ 2. What is the value of $\lim_{x\to0} \frac{\sin(6x)}{\tan(2x)}$?

A) 0
B) $\frac{1}{3}$
C) 3
D) The limit does not exist.

MCQ 3. Let f be the function defined by $f(x)=\frac{3x}{\sqrt{x^2+5}}$. Which of the following statements is true about the horizontal asymptotes of the graph of f ?

A) $y=3$ is the only horizontal asymptote.
B) $y=-3$ is the only horizontal asymptote.
C) $y=3$ and $y=-3$ are both horizontal asymptotes.
D) The graph of f has no horizontal asymptotes.

MCQ 4. The function g is defined by $g(x)=\frac{2|x+3|}{x+3}$ for all $x\neq-3$. What is the value of $\lim_{x\to-3^-} g(x)$?

A) -2
B) 0
C) 2
D) The limit does not exist.

MCQ 5. Let f be the piecewise function defined below.

$$f(x)=\begin{cases}\frac{x^2-25}{x-5}, & \text{for } x\neq 5\\ k, & \text{for } x=5\end{cases}$$

For what value of k is f continuous at $x=5$?

A) 0
B) 5
C) 10
D) 25

Section 2: Free Response Questions

_Show all your work.

**FRQ 1. Let f be the piecewise function defined by:

$$f(x)=\begin{cases}x^2+3x, & \text{for } x<2\\ 10, & \text{for } x=2\\ cx+4, & \text{for } x>2\end{cases}$$

where c is a constant.

(a) Find the value of $\lim_{x\to2^-} f(x)$.

(b) Find the value of c that makes $\lim_{x \to 2} f(x)$ exist. Show the setup for your equation.

(c) Using the value of c found in part (b), determine whether the function f is continuous at $x = 2$. Justify your answer using the three-step definition of continuity.

**FRQ 2. The temperature of water in a tub at time t is modeled by a strictly increasing, continuous function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. Selected values of $W(t)$ are given in the table below for $0 \leq t \leq 20$.

t **(minutes)**	0	4	9	15	20
$W(t)$ (°F)	55	57	62	68	71

Question: Must there be a time t, for $0 < t < 15$, such that the temperature of the water is exactly $65^{\circ}\mathrm{F}$? Justify your answer using a relevant mathematical theorem.